

# Exact Finite-State Coverage of Hellman and Distinguished-Point Time–Memory Tradeoffs: A Reproducible Study on Reduced-DES Mappings

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## Abstract

Cryptanalytic time–memory tradeoffs replace repeated exhaustive search by precomputed chains of an iteration function. Their success depends not on the nominal number of generated chain positions, but on the number of distinct states actually represented after self-intersections and chain merging.

We present a reproducible finite-state study of Hellman and bounded distinguished-point tables for a concrete family of reduced-DES iteration maps. The artifact computes exact state-space coverage and exact key recovery, and provides independent Java and Rust implementations with shared semantic test vectors. In the current  $N = 2^{16}$  configuration with  $m = 1024$  Hellman chains of length  $t = 64$ , the measured median exact coverage is approximately 16.8%. Applying the established random-function coverage recurrence of Ma and Hong predicts approximately 16.651% coverage for the same parameters, showing that the observed loss is consistent with classical random-mapping theory rather than with the independent-sampling heuristic  $1 - e^{-mt/N}$ .

The final version of this report extends this comparison across parameter sweeps, random-function baselines, distinguished-point stopping parameters, and a shared Java/Rust experiment plan. All numerical claims are tied to committed inputs, outputs, and environment metadata.

## 1 Introduction

A time–memory tradeoff (TMTO) moves work from the online phase of a search problem into a reusable preprocessing phase. In Hellman’s classical construction [Hel80], a large key or state space is represented by iteration chains while only chain endpoints are retained. The construction therefore occupies a middle ground between exhaustive search, which uses little memory and large online time, and complete table lookup, which uses large memory and very small online time.

The central implementation issue is that a chain table does not contain  $mt$  independent samples. Once two iterates of the same mapping meet, their future trajectories coincide; a chain can also revisit one of its own previous states. Consequently, the effective coverage of a table is the number of *distinct* represented states, not its nominal number of positions. This phenomenon is part of the classical TMTO literature rather than a new observation: Hellman’s analysis, later work on random mappings, success probability, perfect tables, false alarms, and rainbow reductions all treat different aspects of collisions, merging, and coverage [Hel80, FO90, Oec03, AJO08, MH09, Hon08].

The purpose of this paper is therefore narrower and experimentally explicit. We ask how accurately established random-function TMTO theory describes a *concrete finite cryptographic mapping* when every represented state can be enumerated exactly, and whether independent implementations reproduce the same semantics and finite-state statistics.

**Concrete iteration map.** For a reduced state  $x$ , a canonical DES-key encoding  $K(x)$ , a fixed plaintext  $P$ , encryption  $E$ , and reduction  $R$ , we study

$$F(x) = R(E_{K(x)}(P)).$$

DES is obsolete and is used only as a compact deterministic fixture. The state dimension is deliberately reduced so that exact coverage, cycles, endpoint multiplicities, and recovery behavior can be computed rather than estimated from sampling.

**A motivating discrepancy and its correct baseline.** The repository’s initial  $N = 2^{16}$  experiment used  $m = 1024$  fixed-length chains of length  $t = 64$ , so  $mt = N$ . Treating the  $mt$  chain positions as independent samples would give the occupancy expression

$$1 - e^{-mt/N} = 1 - e^{-1} \approx 63.2\%.$$

That number is *not* the expected coverage of a Hellman matrix. Using the random-function recurrence of Ma and Hong [MH09], the same parameters predict approximately 16.651% coverage of the state space. The measured median exact coverage is approximately 16.8%. Thus the initial measurement is in close agreement with established random-function theory.

This distinction also exposes an important parameterization point. The standard Hellman matrix-stopping regime is expressed in terms of  $mt^2$  relative to  $N$ , not merely  $mt/N$ . Our initial configuration satisfies  $mt = N$  but  $mt^2 = 64N$ ; it should therefore be treated as a finite-state stress point, not as a standard matrix-stopping configuration.

**Contributions and scope.** This paper is an experimental and reproducibility study, not a claim to have discovered chain coalescence or classical TMTO coverage theory. Its intended contributions are:

1. a precise, executable reduced-DES state model with canonical DES parity handling and an explicit reduction;
2. exact coverage and exact-recovery instrumentation for Hellman and bounded distinguished-point tables;
3. direct comparison of concrete cryptographic mappings with established random-function coverage predictions over controlled finite-state parameter sweeps;
4. independent Java and Rust reference implementations, with shared semantic fixtures and, before any timing comparison, a common committed experiment plan;
5. a reproducibility artifact in which numerical claims are generated from committed inputs, outputs, scripts, and environment metadata.

We make no claim to break full-size DES or AES, and no implementation in this repository is intended for production cryptography.

## 2 Related Work

**Classical time–memory tradeoffs.** Hellman introduced the cryptanalytic time–memory tradeoff for block-cipher key recovery and analyzed the effect of chain collisions under random-function

assumptions [Hel80]. Fiat and Naor later gave rigorous time/space tradeoffs for function inversion beyond the random-function setting [FN91]. Barkan, Biham, and Shamir developed a general model of cryptanalytic TMTOs based on stateful random graphs and proved rigorous bounds covering classical and rainbow-style constructions [BBS06]. Recent work continues to revisit the fundamental limits of preprocessing and function inversion [GGPSD23].

**Random mappings and coverage.** Iteration chains naturally induce functional graphs. Flajolet and Odlyzko systematically analyzed random-mapping statistics including cycles, components, and related structural parameters [FO90]. For Hellman tables specifically, Ma and Hong studied the expected coverage rate and supplied a recurrence that closely matches experiments under a random function model [MH09]. Their framework is a central baseline for the present paper: our goal is not to replace it, but to test it against exact coverage of a concrete finite cryptographic mapping.

**Rainbow and perfect tables.** Oechslin’s rainbow construction changes the reduction across chain positions, reducing the conditions under which collisions turn into permanent chain merges [Oec03]. Avoine, Junod, and Oechslin analyzed perfect classic, distinguished-point, and rainbow tables, including practical performance considerations [AJO08]. These works make clear that coverage, preprocessing cost, and lookup cost must be compared at matched success probabilities rather than through nominal table sizes alone.

**False alarms and distinguished points.** Endpoint matches do not always imply successful recovery: false alarms can force regeneration of candidate chains and materially affect the online cost. Hong analyzed this cost for Hellman and rainbow tradeoffs [Hon08]. Lee and Hong compared major perfect-table variants and developed detailed analysis for distinguished-point constructions [LH12].

**Position of this work.** The present paper focuses on reproducibility at finite state sizes where the entire mapping and table coverage can be enumerated. This permits a direct three-way comparison among established random-function predictions, a concrete DES-derived mapping, and independent Java/Rust implementations. The value of the study is therefore in exact instrumentation, controlled comparisons, and reproducible validation—not in reintroducing known TMTO phenomena.

# 3 Model and Definitions

## 4 Random-Function Coverage Baselines

## 5 Reference Implementations

## 6 Experimental Methodology

## 7 Results

## 8 Discussion and Limitations

## 9 Reproducibility

## 10 Conclusion

### References

- [AJO08] Gildas Avoine, Pascal Junod, and Philippe Oechslin. Characterization and improvement of time-memory trade-off based on perfect tables. *ACM Transactions on Information and System Security*, 11(4), 2008.
- [BBS06] Elad Barkan, Eli Biham, and Adi Shamir. Rigorous bounds on cryptanalytic time/memory tradeoffs. In *Advances in Cryptology – CRYPTO 2006*, volume 4117 of *Lecture Notes in Computer Science*, pages 1–21. Springer, 2006.
- [FN91] Amos Fiat and Moni Naor. Rigorous time/space tradeoffs for inverting functions. In *Proceedings of the 23rd Annual ACM Symposium on Theory of Computing*, pages 534–541. ACM, 1991.
- [FO90] Philippe Flajolet and Andrew M. Odlyzko. Random mapping statistics. In *Advances in Cryptology – EUROCRYPT ’89*, volume 434 of *Lecture Notes in Computer Science*, pages 329–354. Springer, 1990.
- [GGPSD23] Alexander Golovnev, Siyao Guo, Spencer Peters, and Noah Stephens-Davidowitz. Revisiting time-space tradeoffs for function inversion. In *Advances in Cryptology – CRYPTO 2023, Part II*, volume 14082 of *Lecture Notes in Computer Science*, pages 453–481. Springer, 2023.
- [Hel80] Martin E. Hellman. A cryptanalytic time–memory trade-off. *IEEE Transactions on Information Theory*, 26(4):401–406, 1980.
- [Hon08] Jin Hong. The cost of false alarms in hellman and rainbow tradeoffs. Cryptology ePrint Archive, Paper 2008/362, 2008.
- [LH12] Ga Won Lee and Jin Hong. A comparison of perfect table cryptanalytic tradeoff algorithms. Cryptology ePrint Archive, Paper 2012/540, 2012.
- [MH09] Daegun Ma and Jin Hong. Success probability of the hellman trade-off. *Information Processing Letters*, 109(7):347–351, 2009.

- [Oec03] Philippe Oechslin. Making a faster cryptanalytic time-memory trade-off. In *Advances in Cryptology – CRYPTO 2003*, pages 617–630. Springer, 2003.